

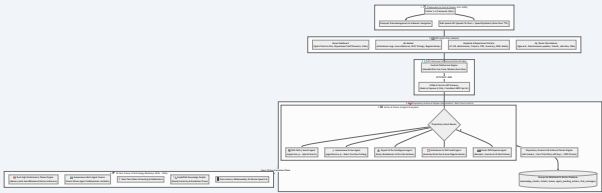
# HRMs-ERP & Proprietary Custom AI Engine

10-Year Architecture Roadmap, Agent Portfolio & Native AI Language Strategy

v3.0.0 Custom AI

100% On-Premise & Independent

## 1. Master System Architecture Chart (Self-Hosted Custom AI)



## 2. AI Agent Portfolio: Active vs. Future Roadmap

✓ AI Agents Created So Far

- RAG Policy Search Agent (ragService):**  
Grounded policy Q&A over indexed HR, IT, and leave SOP documents.
- Autonomous Action Agent (agentService):**  
Converts prompts into backend tool calls (createTicket, applyLeave, sendMessageToUserOrChannel, queryTeamAttendance).
- Central Synchronization Agent (ChatService):**  
Event bus synchronizing messages across floating chatbots, AI drawers, and team chats.
- Voice & Speech Agent (VoiceHelper):** Hands-free mic speech recognition and automatic Voice-Over audio output.
- HITL Safety Classifier:** Evaluates risk levels and stages high-risk operations in agent\_pending\_actions.

🚀 New AI Agents Planned for Development

- Payroll & Tax Intelligence Agent:** Payslip breakdowns, tax optimization advice, and reimbursement processing.
- Attendance & Shift Anomaly Agent:** Automatically detects clock-in anomalies and suggests regularizations.
- Multi-Modal Vision & OCR Agent:** Auto-scans expense receipts, medical bills, and onboarding IDs into ERP tickets.
- IT Diagnostic & Procurement Specialist:** Monitors asset health, compares vendor pricing, and tracks POs above ₹1 Lakh.
- Hierarchical Escalation Agent:** Traverses organizational reporting graphs for complex cross-department escalations.

### 3. Strategic 10-Year Programming Language Evaluation (2026–2036)

To build a proprietary AI engine designed to last for the next 10 years without third-party API dependencies, selecting the right core programming language is critical:

| Language                                                                                                                              | 10-Year Future Suitability          | Strengths for Custom AI Development                                                                                                                                  | Limitations                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
|  <b>Rust</b><br><b>(RECOMMENDED FOR CORE ENGINE)</b> | ★★★★★<br>(Highest Future Proofing)  | Memory safety without Garbage Collection (GC) pauses, zero-cost abstractions, native C/CUDA interop, blazing fast tensor execution (Burn, Candle), WASM compilation. | Steeper learning curve for borrow checker rules.                                              |
|  <b>Python</b>                                       | ★★★★★ (Ideal for R&D / Fine-Tuning) | Unrivalled ecosystem (PyTorch, JAX, HuggingFace, vLLM). Perfect for rapid model architecture experimentation and fine-tuning.                                        | Higher runtime memory footprint, GIL overhead, slower execution latency without C++ bindings. |
|  <b>C++ / CUDA</b>                                  | ★★★★★<br>(Low-Level GPU Kernels)    | Foundation of GGML, llama.cpp, and TensorRT. Maximum low-level GPU hardware control.                                                                                 | Manual memory management risks, complex cross-platform maintenance.                           |
|  <b>Mojo / Julia</b>                               | ★★★<br>(Emerging Contenders)        | Mojo compiles Python syntax to native C-speed code. High-performance matrix math capabilities.                                                                       | Ecosystem still maturing compared to Rust/Python.                                             |

**Strategic 10-Year Architecture Recommendation: A Hybrid Microservice Architecture:**

- **Rust (or C++/CUDA):** Core high-throughput, low-latency Custom AI Inference Engine & Tensor Operations.
- **Python:** Model Training, Dataset Preparation & Fine-Tuning (LoRA).
- **Node.js / Express:** API Gateway & Application Routing (Port 4001).
- **Dart (Flutter):** Cross-Platform Client UI for Web, Mobile & Desktop (Port 4000).